Analyzing the Impact of Economic Factors on Housing Prices in the United States.

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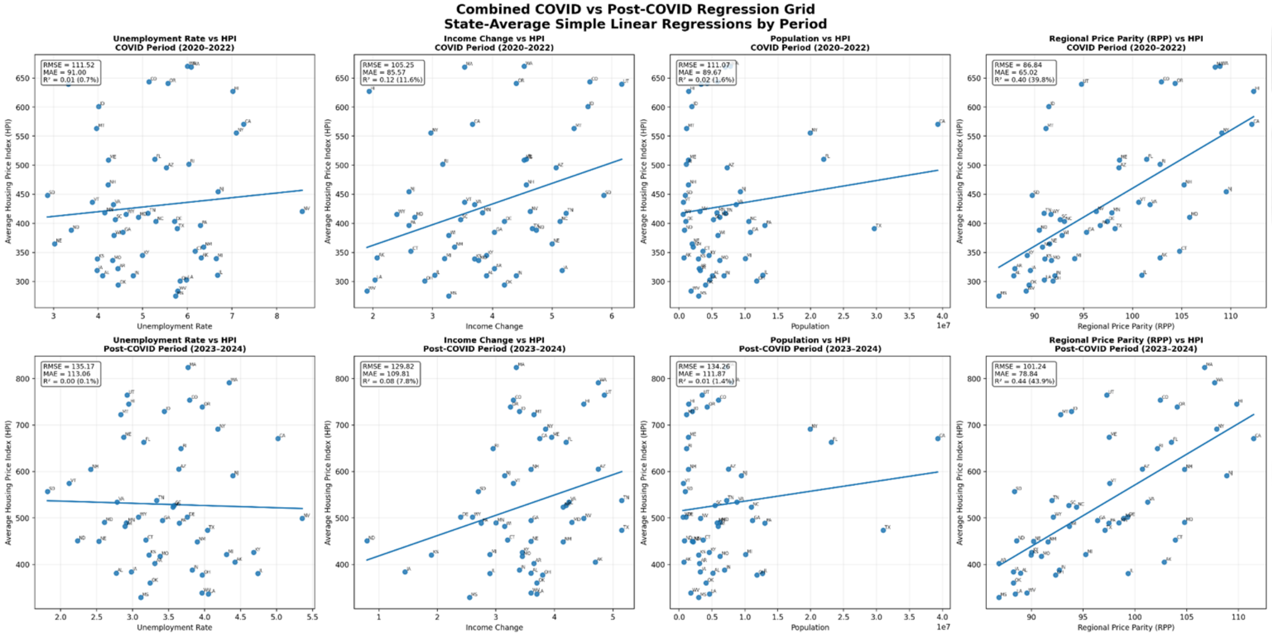
Problem Statement

- Housing prices affect affordability, migration, and financial stability.
- Economic conditions vary widely across U.S. states.
- States differ in income growth, unemployment, population change, migration, and cost of living.
- Research question: How are these factors associated with housing prices from 2020 to 2024?
- Main goal: explain housing prices using a multivariable approach.

Exploratory Data Analysis(EDA)

- Housing prices increased during both COVID and post-COVID periods.
- RPP showed the strongest positive relationship with housing prices.
- Unemployment showed a negative relationship.
- Income change showed a moderate positive relationship.
- Relationships varied slightly between COVID and post-COVID periods.

Regression Grid



HPI: Housing Price Index measuring relative housing-price movement over time.

RPP: Regional Price Parity measuring state cost-of-living differences.

This regression grid compares several economic variables against housing prices during both the COVID and post-COVID periods.

The charts show that RPP had the clearest positive relationship with housing prices, while unemployment generally showed a weaker negative relationship.

Income change showed a moderate positive trend, while population effects were smaller.

The grid also helps visualize how these relationships changed slightly between the COVID and post-COVID periods.

Proposed Solution

Combined yearly data from all 50 states from 2020–2024 into a panel dataset.

Used a two-way fixed effects panel regression as the main model.

Used Random Forest Regression as a comparison model.

Compared economic relationships during COVID and post-COVID periods.

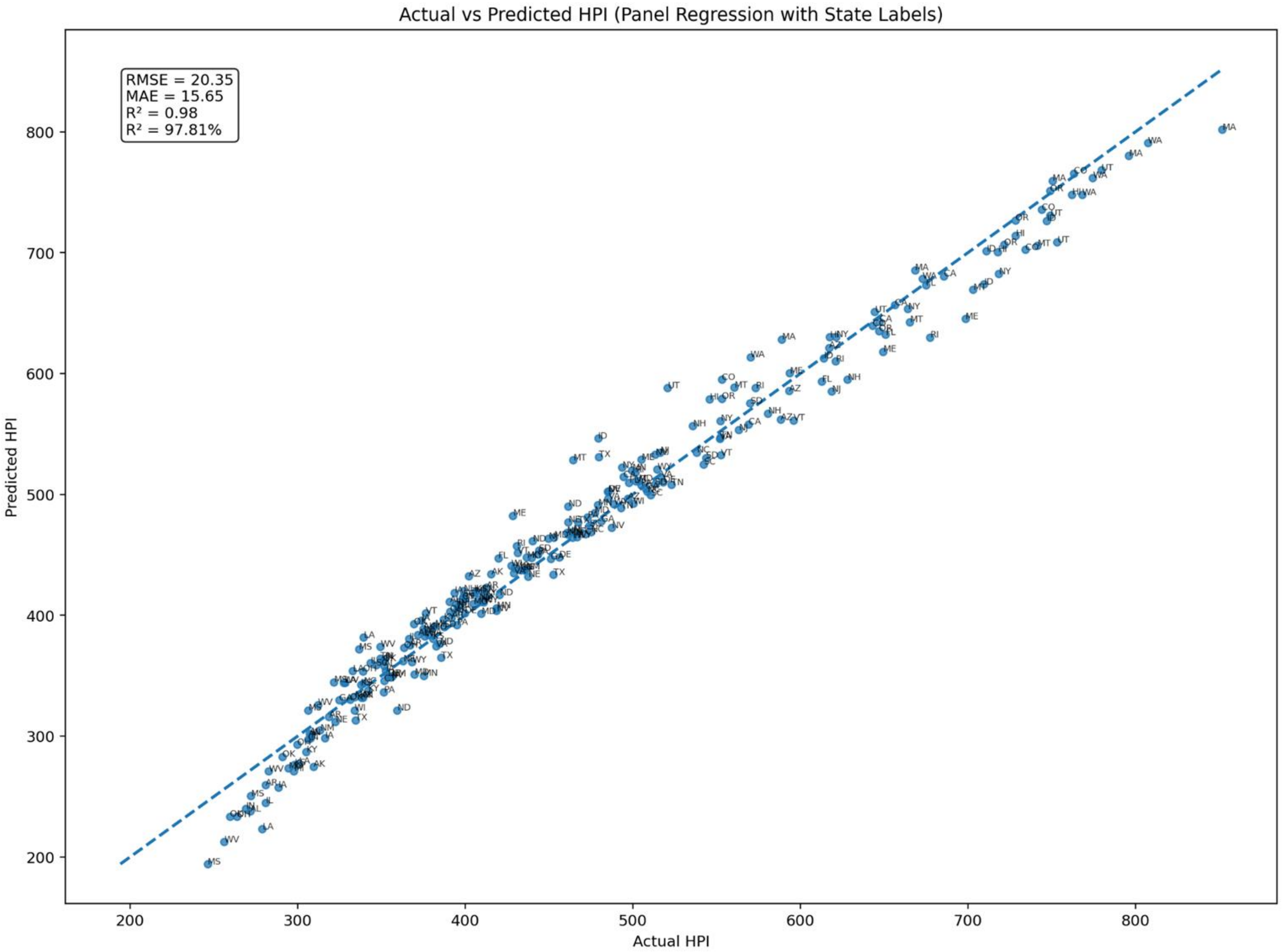
Controlled for both state and year effects.

Housing Prices

Panel regression was appropriate because the same states were observed across multiple years.

Model Prediction Performance

- The two-way fixed effects panel regression explained a large share of housing-price variation across the 50-state panel from 2020 to 2024.
- RPP remained positively associated with housing prices.
- Unemployment remained negatively associated with housing prices over time.
- Population had a weaker effect than the main labor and cost-of-living variables.
- The model fit the data strongly, with R^2 = 97.81%, RMSE = 20.35, and MAE = 15.65.



The Actual vs Predicted HPI chart shows that most points fall close to the diagonal line, indicating that the panel model tracked housing-price patterns well.

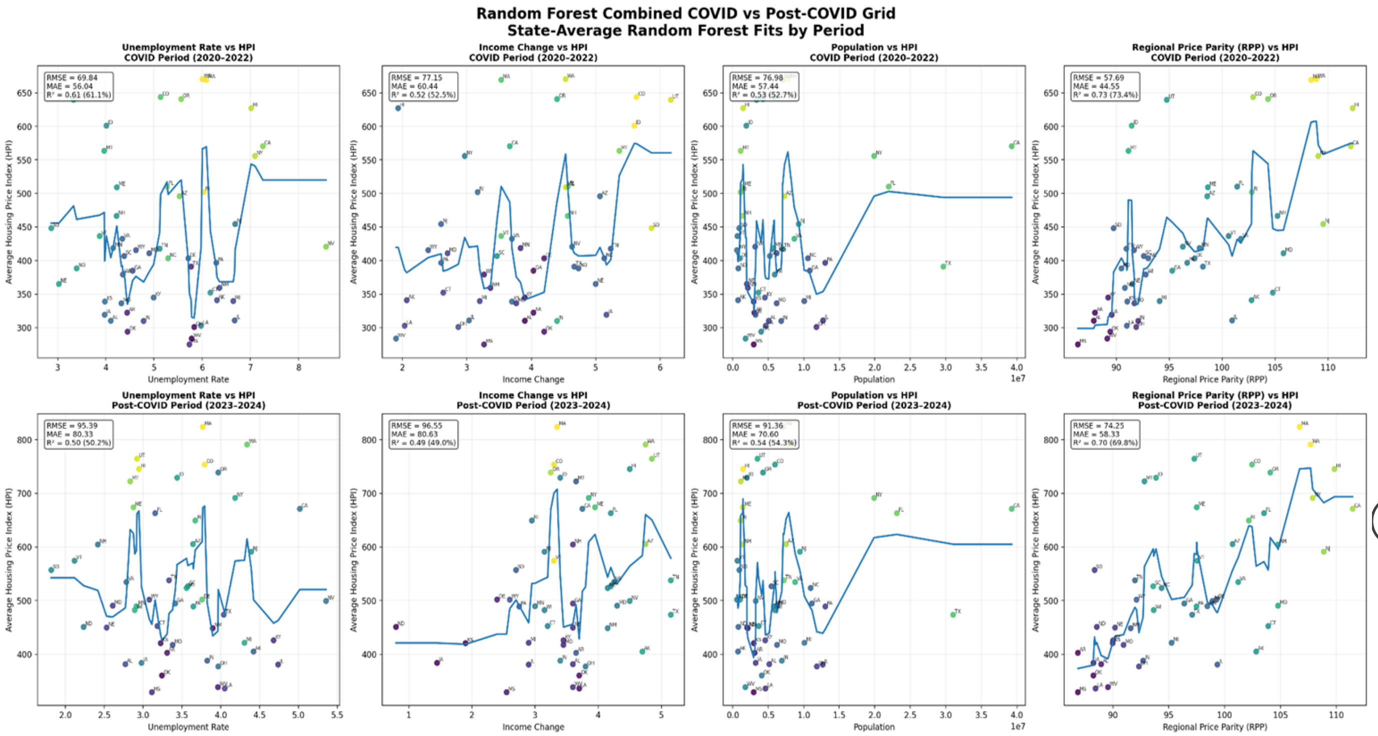
This chart compares actual housing prices with the panel model’s predicted values.

The dashed diagonal line represents perfect predictions.

Points closer to the line indicate better model fit.

Overall, the panel regression tracked housing-price patterns well across states.

Random Forest Comparison



- Random Forest supported similar relationships found in panel regression.
- RPP remained the strongest positive predictor.
- Unemployment showed weaker negative patterns.
- Relationships varied between COVID and post-COVID periods.
- Random Forest captured more complex non-linear relationships.

This grid shows Random Forest relationships between economic variables and housing prices during the COVID and post-COVID periods.

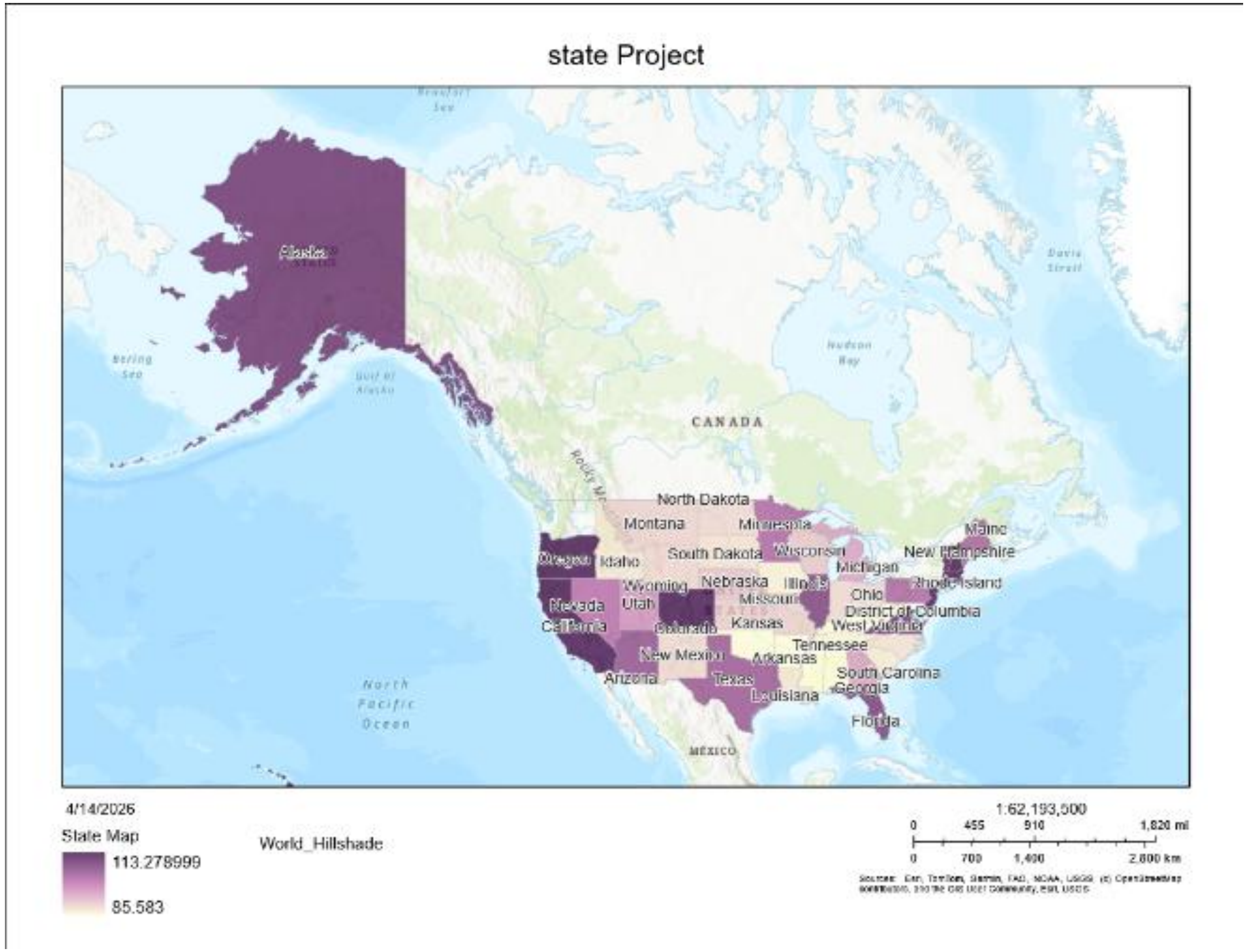
Similar to the panel regression results, RPP remained the strongest positive relationship with housing prices.

Unemployment showed weaker negative trends, while income changes showed moderate positive relationships.

Random Forest was useful because it captured more complex and non-linear patterns that linear regression may miss.

Choropleth Map

- Darker states represent higher housing price levels.
- Coastal and higher-cost states generally showed higher HPI values.
- The map highlights geographic housing-price disparities .



The choropleth map shows regional housing-price differences across the United States.Higher-cost coastal states generally showed higher housing-price levels.

Conclusion And future Work

- Housing prices are influenced by multiple economic factors.
- RPP remained one of the strongest and most consistent factors
- Unemployment remained negatively associated with housing prices.
- Model performance differed across states.
- Future work could include mortgage rates, housing supply, and additional fine tuning.

References

- U.S. Census Bureau.
- Bureau of Labor Statistics.
- Federal Housing Finance Agency.
- Bureau of Economic Analysis.
- IRS migration data.